

The Attention-Semantic Interface: A Neurocognitive Framework for the Dynamic Interplay of Meaning and Control

Part I: The Neural Architecture of Semantic Cognition

Semantic cognition, the faculty that allows us to access and manipulate our vast store of general knowledge about the world, is not the product of a single, isolated brain region. Rather, it is an emergent property of a large-scale, distributed neural system. Understanding the architecture of this system—its constituent nodes, the connections that bind them, and its principles of organization—is a prerequisite for comprehending its intricate relationship with attentional control. Decades of research, leveraging evidence from healthy cognition, brain damage, and advanced neuroimaging, have revealed a system that is both widely distributed and functionally specialized, organized hierarchically to support the flexible use of meaning.¹

A Distributed Network for Meaning: Nodes and Connectomes

The foundation of semantic processing is a network of anatomically distinct gray matter (GM) regions, or "nodes," primarily located within the temporal, parietal, and frontal lobes. These nodes are structurally interconnected by a complex web of white matter (WM) tracts, forming a "semantic connectome" that facilitates the communication necessary for conceptual thought.¹ Functional magnetic resonance imaging (fMRI) and positron emission tomography (PET) studies consistently identify a core set of regions that activate during tasks requiring access to the meaning of words, pictures, objects, and sounds.

Key GM nodes of this canonical semantic network include several critical territories. Within the temporal lobe, the **anterior temporal lobe (ATL)** has emerged as a region

of paramount importance.² This is complemented by significant involvement of the **middle and inferior temporal gyri (MTG/ITG)**, the **fusiform gyrus**, and the **superior temporal gyrus (STG)**.² In the parietal lobe, the **angular gyrus (AG)**, often considered part of the **temporoparietal junction (TPJ)**, and the **precuneus/posterior cingulate cortex (PCC)** are consistently implicated.² The frontal lobe contributes crucial control and integration functions via the **inferior frontal gyrus (IFG)**, particularly its anterior portions often associated with Broca's area, as well as the **dorsomedial and medial prefrontal cortex (DMPFC/mPFC)**.² Finally, limbic structures such as the **hippocampus** are integral to the network, playing a vital role in the interface between semantic and episodic memory, a link made starkly clear in neurodegenerative conditions like semantic dementia.⁶

The functional cooperation of these disparate GM nodes is contingent upon the structural integrity of the WM pathways that connect them. The development of diffusion imaging techniques, such as diffusion tensor imaging (DTI), combined with tractography, has allowed for the in vivo mapping of these anatomical connections. Crucially, by correlating measures of tract integrity—such as fractional anisotropy (FA) or the extent of lesion damage—with semantic performance in brain-damaged patients, researchers have moved beyond mere anatomical description to establish a causal link between specific pathways and semantic ability.¹ This lesion-symptom mapping approach has identified several major WM fasciculi as forming the backbone of the semantic connectome. These include the left-lateralized

inferior fronto-occipital fasciculus (IFOF), which links frontal, temporal, and occipital regions; the **uncinate fasciculus (UF)**, connecting the anterior temporal lobe with the orbitofrontal cortex; and various components of the **superior longitudinal fasciculus (SLF)**, which provide critical fronto-parietal and fronto-temporal connectivity.¹ The finding that damage to these specific tracts, independent of damage to the GM nodes they connect, predicts semantic deficits underscores that semantic cognition is fundamentally a network-level phenomenon, dependent on the brain's structural wiring.¹

The Hub-and-Spoke Model: A Principle of Semantic Organization

To organize this complex, distributed network, a powerful theoretical framework has been proposed: the "hub-and-spoke" model.¹⁰ This model posits a functional hierarchy within the semantic system. It suggests that modality-specific sensory, motor, and affective information, which constitutes our primary experience of the world, is stored in disparate cortical regions known as "spokes." For instance, visual features of an object are processed in the visual cortex, its associated actions in the motor cortex, and its typical sounds in the auditory cortex. The model proposes that a transmodal "hub" is necessary to integrate this distributed information into coherent, amodal conceptual representations that are abstracted away from any single sensory modality.²

A wealth of convergent evidence points to the bilateral anterior temporal lobes (ATLs) as the principal neuroanatomical substrate for this semantic hub.² Evidence for this central role comes from multiple lines of inquiry. First, functional connectivity studies using resting-state fMRI, which measures spontaneous, correlated fluctuations in brain activity, consistently demonstrate that the ATLs maintain strong intrinsic connections with the other distributed nodes of the semantic network, including the IFG, AG, and mPFC.² This pattern of connectivity is not limited to the resting state; psychophysiological interaction (PPI) analyses show that this same core network is functionally coupled during active, effortful semantic tasks.² The existence of this stable "default-semantic" network, active both during internally-directed thought and externally-cued tasks, suggests that the ATL's integrative function is a continuous and fundamental aspect of cognition.²

Second, and perhaps most compellingly, neuropsychological studies of patients with semantic dementia (SD) provide causal evidence for the ATL's role. SD is a neurodegenerative disorder defined by the progressive and profound loss of conceptual knowledge, affecting all modalities of input (words, pictures, sounds) and output (naming, drawing, object use).⁶ This clinical syndrome is robustly associated with focal, often asymmetric, atrophy of the ATLs.³ The direct correspondence between the degradation of the ATL tissue and the dissolution of semantic memory provides powerful support for the hypothesis that the structural integrity of this region is necessary for the formation and maintenance of amodal conceptual representations.

Further research has revealed that the ATL hub is not a functionally uniform entity. There appears to be a graded specialization within the ATL itself. Using seed-based connectivity analyses, studies have shown that the ventrolateral aspects of the ATL,

including the ventral ATL and anterior middle temporal gyrus (aMTG), exhibit the strongest connectivity with the core multimodal semantic network. In contrast, more superior regions, such as the anterior superior temporal gyrus and superior temporal sulcus (aSTG/STS), are more tightly coupled with auditory and language-specific processing areas.² This suggests a fine-grained functional topography within the hub, with different subregions potentially mediating different aspects of semantic integration, moving from more modality-specific (e.g., auditory-linguistic) to fully transmodal representations.

Functional Specialization: Social Semantics and Conceptual Accumulation

Beyond the overarching hub-and-spoke architecture, the semantic system exhibits further functional specialization, with evidence for subsystems that are preferentially tuned to specific types of conceptual content.⁸ One of the most well-documented examples is the "social semantic network." This is a specific constellation of brain regions, largely overlapping with the core semantic network and the default mode network (DMN), that shows selective activation when processing social concepts, such as personality traits, social situations, and the mental states of others. This network consistently engages the bilateral ATLs, TPJ/AG, DMPFC, and the posterior cingulate/precuneus.⁸

Critically, this network does more than simply represent the meaning of individual words with social connotations. It supports the dynamic, online accumulation and integration of connected meaning as it unfolds in sentences and narratives.⁸ This process of semantic accumulation reveals that comprehension is not a static, word-by-word lookup process. fMRI studies have demonstrated that the activation of the social semantic network is better explained by the holistic social-semantic richness of an entire sentence or narrative than by the summed semantic content of its constituent words.⁸ This finding indicates a true integration of meaning over time, where the network constructs and updates a coherent representation of the unfolding social context. This effect is semantic, not syntactic, as it disappears when content words are replaced with pseudowords.⁸ This dynamic integration process is a key feature of how the brain derives complex meaning from language, moving beyond individual concepts to build rich situational models.

The principle of functional specialization extends beyond the social domain. A more complex "semantic atlas" is emerging, suggesting that the brain organizes conceptual

knowledge along multiple fundamental dimensions, including not only socialness but also sensory-motor attributes (vision, action), emotion, space, and time.¹³ Brain regions that are selectively sensitive to these different dimensions are found both within and outside the canonical semantic network. This points to a highly nuanced organization where different aspects of a concept's meaning are rooted in different neural systems. Notably, certain regions within the Default Mode Network, particularly the left aSTG and AG, appear to function as "connector hubs" where the neural correlates of these disparate semantic dimensions overlap and are integrated.¹³

This evidence collectively refines our understanding of the semantic system, moving it beyond a simple, undifferentiated network. The system appears to be organized according to a multi-layered, hierarchical architecture. At the highest level, the hub-and-spoke model describes the integration of modality-specific information into amodal concepts within the ATL. Within this framework, however, there are functionally specialized subsystems, like the social semantic network, that are optimized for processing particular domains of knowledge. Furthermore, the process of comprehension itself is not static but dynamic, involving the continuous accumulation and integration of meaning over time. This dynamic integration of complex, abstract information finds its neural substrate in the high-level association cortices that constitute the Default Mode Network. The DMN, traditionally viewed as a "task-negative" or "rest" network, is increasingly understood to provide the core associative architecture for sophisticated semantic cognition, a theme that will be central to understanding its interface with attention.

Part II: The Neurobiology of Attentional Control

Attention is not a single entity but a collection of cognitive processes that enable the brain to selectively manage information in a world of overwhelming sensory input. Given its finite processing capacity, the brain requires mechanisms to prioritize neural representations that are most relevant to current behavioral goals.¹⁵ This selective prioritization, which we call attention, is a cornerstone of cognition, influencing perception, memory, and action. It is subserved by a set of large-scale brain networks whose neurobiology is shaped by genetic, environmental, and developmental factors.¹⁵

Fundamental Principles and Biological Foundations of Attention

At a conceptual level, attention is often parsed into several key dichotomies that reflect its different forms and triggers.¹⁹ The most fundamental of these is the distinction between two modes of control.

Top-down attention, also known as endogenous or voluntary attention, refers to the goal-directed allocation of processing resources. This is what occurs when one deliberately searches for a friend's face in a crowd or focuses on a specific instrument in an orchestra. This form of attention is typically slow, effortful, and under conscious control.²⁰ In contrast,

bottom-up attention, also known as exogenous or stimulus-driven attention, is the involuntary and rapid capture of processing resources by a salient or unexpected event in the environment, such as a sudden flash of light or a loud bang. This mechanism is fast, automatic, and serves to alert the organism to potentially important changes in its surroundings.¹⁹ Other distinctions include overt versus covert attention (whether attention is accompanied by a physical movement of a sensory organ, like the eyes) and selective versus nonselective attention (the latter relating more to general arousal or vigilance).¹⁹

The development and function of these attentional capacities are rooted in a complex interplay of biological factors. Genetic predispositions play a significant role, with specific gene variants influencing attentional abilities. For example, genes involved in the regulation of key neurotransmitter systems, such as the dopamine receptor D4 gene (DRD4) and the catechol-O-methyltransferase (COMT) gene, have been linked to individual differences in attentional processing.¹⁵ These genetic factors interact with environmental influences, such as family environment and cultural context, to shape attentional development.¹⁵ At the neural level, the maturation of attentional networks relies on fundamental processes of brain development.

Synaptic plasticity, the ability of neural connections to strengthen or weaken based on activity, allows for the refinement of attentional circuits over time.¹⁵ This is complemented by

neural pruning, the elimination of redundant or inefficient connections, and **myelination**, the insulation of nerve fibers that speeds up the transmission of neural signals. Together, these processes sculpt efficient and highly specialized attentional

networks.¹⁵

The Dual-Network Model of Visual Attention

The conceptual distinction between top-down and bottom-up attention has a clear and robust neuroanatomical correlate in the form of two large-scale, functionally distinct brain networks.²⁰ This dual-network model, primarily elucidated in the domain of visual attention, provides a powerful framework for understanding the neural basis of attentional control.

The **Dorsal Attention Network (DAN)** is a bilateral system with its core nodes located in the **intraparietal sulcus (IPS)** and the **frontal eye fields (FEF)**.²⁰ The DAN is the primary neural substrate for

top-down, goal-directed attention. It becomes active when an individual voluntarily orients their attention to a specific spatial location or visual feature, for instance, in response to a predictive cue.²⁴ The DAN is responsible for maintaining task goals and exerting top-down biasing signals that modulate activity in sensory cortices. For example, when attending to a specific location, the DAN enhances the neural representation of stimuli appearing at that location in the visual cortex, effectively turning up the "gain" on relevant information.²⁰

The **Ventral Attention Network (VAN)**, in contrast, is a strongly right-lateralized system anchored by the **temporoparietal junction (TPJ)** and the **ventral frontal cortex (VFC)**.²⁰ The VAN is the primary system for

bottom-up, stimulus-driven attention. Its core function is to act as a "circuit-breaker" for the goal-directed processing of the DAN.²⁰ The VAN is activated by behaviorally relevant but unexpected or infrequent stimuli, particularly those appearing outside the current focus of attention. Upon detecting such a stimulus, the VAN interrupts the DAN's ongoing top-down control, triggering a reorienting of attention toward the new, salient event.²⁰

Dynamic Interaction and Network Integration

Crucially, flexible and adaptive attentional control does not arise from the isolated action of either the DAN or the VAN. Instead, it emerges from their continuous and dynamic interaction.²³ The functional relationship between these two networks is not fixed but is highly dependent on the cognitive context and current task demands. For example, during tasks that require sustained, focused top-down attention (like visual search for a difficult target), the DAN is highly active while activity in the VAN, particularly the TPJ, is actively suppressed.²⁴ This suppression is thought to be a filtering mechanism, initiated by the DAN, to protect goal-directed behavior from being derailed by irrelevant distractors. Conversely, when a salient, unexpected target appears and requires a shift of focus, the two networks become co-activated and functionally coupled to execute the reorienting of attention.²³

This functional interplay is supported by a specific structural architecture. While the DAN and VAN are largely segregated networks, they are linked by major WM pathways, most notably the **superior longitudinal fasciculus (SLF)**. Different branches of this large tract serve to connect nodes both within each network (e.g., SLF I connecting dorsal frontoparietal areas) and, critically, between the two networks (e.g., SLF II connecting the parietal component of the VAN with prefrontal nodes of the DAN), providing the anatomical substrate for their interaction.²³

The profound interdependence of these networks is vividly illustrated by clinical syndromes such as spatial neglect. This condition, which involves a failure to attend to stimuli on one side of space, is often caused by damage to nodes of the right-hemisphere VAN, such as the right TPJ. However, the functional consequences are not limited to the damaged network; patients with neglect also show abnormal patterns of activity and connectivity in their structurally intact DAN.²⁴ This demonstrates that a lesion in the bottom-up "circuit-breaking" system effectively cripples the top-down control system, providing powerful causal evidence that the two networks must work in concert for normal attentional function.

This dual-network architecture can be understood as a fundamental evolutionary solution to a critical survival problem: how can an organism maintain focus on a current goal (e.g., foraging for a specific type of food) while simultaneously remaining vigilant to unexpected threats or opportunities in the environment (e.g., a predator or a more valuable food source)? The DAN provides the mechanism for sustained, goal-directed focus, while the VAN provides the essential, life-saving interrupt signal. Their dynamic interplay allows for a flexible balance between stability and vigilance. This network-based understanding of attention also provides a modern lens through which to view clinical disorders. Conditions like ADHD, once conceptualized in terms of deficits in a single brain region like the prefrontal cortex, are now increasingly seen

as disorders of large-scale network dynamics and connectivity, reflecting a breakdown in the coordinated activity of these distributed attentional systems.²⁶

To consolidate the key neural players before examining their interaction with semantic systems, the following table summarizes the primary roles of the core regions discussed.

Region	Abbreviation	Primary Network(s)	Core Function(s)
Anterior Temporal Lobe	ATL	Semantic Hub, DMN-FT	Amodal conceptual representation; integration of sensory information.
Angular Gyrus	AG	Semantic, DMN-Core	Disputed: Semantic integration, episodic retrieval, or domain-general buffer.
Inferior Frontal Gyrus	IFG	Semantic Control, DMN-FT	Controlled retrieval, selection, and unification of semantic representations.
Intraparietal Sulcus	IPS	DAN	Top-down spatial/feature-based attentional orienting; maintaining spatial priority maps.
Frontal Eye Fields	FEF	DAN	Top-down control of eye movements and spatial attention.
Temporoparietal Junction	TPJ	VAN, Social Semantic	Bottom-up detection of salient, unexpected events; reorienting attention.
Ventral Frontal Cortex	VFC	VAN	Part of the bottom-up

			"circuit-breaking" system.
Medial Prefrontal Cortex	mPFC	DMN-Core, Semantic	Self-referential processing; social semantic processing; context encoding.

Part III: The Dynamic Interplay of Semantics and Attention

Having established the distinct neural architectures for semantic cognition and attentional control, the central task is to understand how these systems interact. This interaction is not a one-way street but a dynamic, bidirectional loop: goal-directed attention actively shapes how we access and use our knowledge, while the semantic content of our environment powerfully and automatically guides where we focus our attention. This continuous interplay is fundamental to flexible, intelligent behavior, allowing us to navigate a complex world by seamlessly integrating our internal goals with the meaning latent in our sensory inputs.

Top-Down Modulation: How Attention Shapes Semantic Retrieval

The vastness of our semantic memory necessitates a control mechanism. We cannot, and do not, activate all possible knowledge associated with a concept at once. Instead, goal-directed attention provides top-down control over semantic retrieval, allowing us to flexibly navigate our conceptual knowledge base to select information that is relevant to the current context or task.²⁸ This modulation is an active process of biasing competition: attentional signals enhance the neural representation of task-relevant semantic features while simultaneously suppressing irrelevant or distracting ones.²⁸

The neural mechanism for this top-down modulation involves the engagement of a network of prefrontal and parietal control regions—including nodes of the DAN and a more specialized semantic control network—which transmit biasing signals to the

temporal lobe regions that house semantic representations.²² This process is particularly evident in situations of semantic conflict. The classic Stroop task, in which participants must name the ink color of a word while ignoring the word's meaning (e.g., the word "RED" printed in blue ink), provides a clear example. To succeed, participants must exert attentional control to inhibit the highly automatic and prepotent pathway of reading the word and instead prioritize the less automatic task of naming the color.³² Neuroimaging studies of such tasks reveal that conditions of high semantic conflict are associated with increased functional connectivity between key control regions, like the inferior frontal gyrus (IFG), and semantic hubs, like the anterior temporal lobe (ATL). This heightened coupling reflects the increased demand for top-down modulation to resolve the conflict between competing semantic and perceptual pathways.³⁵

The influence of attention extends to more subtle semantic processes like priming. Semantic priming—the phenomenon where processing a target (e.g., "doctor") is facilitated by prior exposure to a related prime (e.g., "nurse")—is not a purely automatic process. Its magnitude is significantly modulated by attentional control. When experimental conditions allow participants to form conscious expectancies about an upcoming target, the resulting priming effect is larger and more robust, a process that consumes attentional resources.³⁶ Multimodal neuroimaging, which combines the temporal precision of EEG/MEG with the spatial resolution of fMRI, has been instrumental in dissociating the different stages of this process. These studies suggest that an early, largely automatic component of associative priming occurs within the anterior temporal cortex between 300 and 500 milliseconds post-stimulus. This is often followed by later-stage activity in frontal and parietal control regions, which may reflect more conscious, attention-mediated processes of retrospective matching or evaluation of the prime-target relationship.³⁷

Semantic Guidance: How Meaning Directs Attentional Focus

The flow of influence between attention and semantics is bidirectional. Just as top-down attention shapes semantic processing, the semantic content of the environment provides a powerful, often automatic, set of signals that guide the deployment of attention. This "semantic guidance" demonstrates that where we look is determined not just by what is visually salient (e.g., bright or colorful) but, more profoundly, by what is meaningful.³⁸

Several mechanisms underlie this semantic guidance of attention. First, the **gist** of a scene—its basic meaning or category (e.g., "kitchen," "beach")—is extracted with remarkable speed, often within the first 100 milliseconds of viewing. This rapidly-perceived context immediately constrains attentional allocation, guiding our eyes toward locations where objects are likely to be found.³⁸ Second, once objects are processed, their relationship to the scene context influences attention.

Semantically inconsistent objects (e.g., a typewriter in a forest) and **spatially inconsistent objects** (e.g., a fire hydrant on a roof) violate our expectations and rapidly attract fixations, suggesting that the brain continuously evaluates the coherence of the visual world.³⁸ Third, the relationships between objects themselves guide attention. During free viewing, our gaze tends to jump between

semantically related objects (e.g., from a coffee cup to a coffee pot). During visual search, our attention is automatically biased toward objects that are conceptually similar to the search target, even if they are not the target itself.³⁸

The potency of semantic guidance is striking. In studies that directly pit the influence of low-level visual salience against high-level meaning by creating "meaning maps" and "saliency maps" of scenes, meaning consistently emerges as the dominant factor in predicting where people look. This holds true even in tasks where meaning is entirely irrelevant to the participant's goal, such as judging the brightness of pixels.⁴⁰ This suggests that the extraction of meaning and its use to guide attention is a fundamental and often obligatory aspect of visual processing.

This evidence paints a picture of the attention-semantic relationship as a continuous, recursive control loop. Attention is not a simple "spotlight" that passively illuminates a static map of knowledge. Instead, attentional control systems (like the DAN and a specialized semantic control network) actively sculpt and constrain the process of semantic retrieval to suit our goals. Simultaneously, the semantic system is constantly and automatically processing the meaning of the world around us, and this extracted meaning, in turn, provides a powerful set of priors that guide where the attentional "spotlight" will be aimed next. This bidirectional loop is the essence of flexible, goal-directed cognition.

The Controlled Semantic Cognition (CSC) Framework: An Integrative Model

To formalize the dynamic interplay between knowledge representation and executive control, a comprehensive neurocognitive model known as the Controlled Semantic Cognition (CSC) framework has been proposed by Jefferies, Lambon Ralph, and colleagues.¹⁰ This framework provides a powerful synthesis of the evidence from neuroimaging, neuropsychology, and neurostimulation, and offers a mechanistic account of how the brain achieves flexible, context-appropriate semantic thought.

The CSC framework is built upon the interaction of two principal, large-scale neural systems:

1. **The Semantic Representation System:** This is the "hub-and-spoke" network detailed in Part I. It is responsible for storing our amodal conceptual knowledge. The anterior temporal lobe (ATL) serves as the transmodal hub, integrating information from modality-specific "spokes" distributed throughout the cortex.¹⁰ This system forms the long-term store of our world knowledge.
2. **The Semantic Control Network (SCN):** This is a functionally distinct frontoparietal network that is responsible for the top-down regulation of semantic processing. Its core nodes include the **left inferior frontal gyrus (IFG)**, the **posterior middle temporal gyrus (pMTG)**, and the **dorsomedial prefrontal cortex (DMPFC)**.⁴² The SCN is not involved in storing knowledge itself, but rather in shaping its retrieval and application. Its role becomes critical when tasks require us to move beyond dominant, automatic associations and retrieve weaker or less common aspects of a concept's meaning, or to resolve competition between multiple activated concepts.⁴¹

The central tenet of the CSC model is the **dynamic, graded interaction** between these two systems. The degree to which the SCN is engaged depends entirely on the cognitive demands of the current task.

- **Automatic Semantic Processing:** When a task involves strong, highly familiar, and dominant semantic associations (e.g., retrieving "bark" for the concept DOG), the process is relatively automatic. It relies primarily on the spreading of activation within the representation system (the hub-and-spoke network), requiring minimal top-down input from the SCN.¹⁰
- **Controlled Semantic Processing:** In contrast, when a task is more demanding, the SCN "ramps up" its activity to guide retrieval. This occurs in several key situations: when we need to retrieve a weak or non-dominant association (e.g., retrieving "pet" for DOG instead of the more dominant "bark"); when we need to resolve high competition from related concepts (e.g., selecting "cat" from among other pets); or when we need to use knowledge in an atypical way (e.g., judging objects based on their color rather than their function).¹⁰ In these situations of

high control demand, the SCN not only increases its own activity but also dynamically increases its functional connectivity with the relevant parts of the representation system—either the ATL hub for amodal concepts or specific spokes for modality-specific features—to actively shape the pattern of semantic activation.¹¹

This framework moves beyond a simple dichotomy of "representation vs. control" to a more nuanced, interactive model. It suggests that the brain possesses a control system specifically tailored for the unique challenges of navigating the vast and complex space of semantic memory. This Semantic Control Network is functionally dissociable from the domain-general Multiple-Demand Network (MDN), which is engaged by difficult tasks of all kinds, not just semantic ones.⁴² This specialization implies that managing conceptual knowledge may require a unique set of computational mechanisms, distinct from those needed for other executive functions like spatial working memory. The CSC framework thus provides a detailed, neurally-plausible account of how the brain achieves the remarkable flexibility that characterizes human thought.

The Role of the Default Mode Network (DMN) at the Interface

The Default Mode Network (DMN) occupies a critical and contentious position at the interface of attention and semantics. The DMN is a set of brain regions—prominently including the posterior cingulate cortex (PCC), medial prefrontal cortex (mPFC), angular gyrus (AG), and medial temporal lobes—that are characterized by high metabolic activity at rest and consistent deactivation during many externally-focused, attention-demanding tasks.⁴⁴ This "task-negative" behavior led to initial theories linking the DMN to internal, self-referential processes like mind-wandering and autobiographical memory retrieval.⁴⁴ However, a paradox emerged: many core DMN nodes are also consistently activated during demanding semantic tasks.⁵ The "semantic hypothesis" of the DMN attempts to resolve this by proposing that the network's baseline activity reflects the continuous, spontaneous flow of conceptual thought that occurs during "rest".⁴⁸

A resolution to this paradox is emerging from two key lines of evidence. First, the DMN is not a monolithic entity but is composed of functionally distinct subsystems.⁴⁵ These include:

- A **Core Subsystem**, centered on the PCC and mPFC, which is most strongly linked to self-referential cognition and shows the most consistent task-induced deactivation.
- A **Medial Temporal Subsystem**, including the hippocampus, which is primarily involved in episodic memory and constructing mental scenes.
- A **Frontotemporal (FT) or Dorsomedial Prefrontal (dmPFC) Subsystem**, which includes the dmPFC, IFG, and ATL. This subsystem is strongly implicated in semantic processing. Critically, its activity has been shown to *facilitate*, not hinder, successful memory encoding for verbal information, directly challenging the notion that all DMN activity is detrimental to task performance.⁴⁵

Second, the DMN is not a static network; its functional configuration is dynamic and task-dependent. Key regions can effectively "switch allegiance" between networks depending on the cognitive context. For example, the ATL and AG, often considered parts of the DMN, behave as such during non-semantic tasks (i.e., they deactivate). However, during semantic tasks, they activate and behave as core nodes of the semantic network.⁴⁸

This evidence forces a reconceptualization of the DMN. It is not simply a "rest network" or a "semantic network." Rather, it appears to be a flexible, high-level cognitive workspace composed of the brain's most abstract, heteromodal association cortices. Its subsystems can be flexibly recruited and configured to support different forms of context-rich, model-based cognition. When the cognitive context is internally generated—such as recalling a past event or imagining the future—the DMN subsystems work in concert to construct and manipulate these mental scenes. When the context is external but conceptually deep and demanding—such as comprehending a complex narrative—the frontotemporal subsystem is engaged to support the necessary semantic processing. The DMN is thus best understood not by a single function, but as the brain's primary engine for creating and navigating the rich, integrated mental models that form the basis of both internal thought and deep understanding of the external world.

Part IV: Clinical Perspectives: Dissociations and Dysfunctions at the Attention-Semantic Interface

The study of neuropsychological disorders offers a powerful lens through which to

examine the functional architecture of the brain. By observing the specific cognitive deficits that result from damage to particular neural systems, we can move from correlational evidence provided by neuroimaging to causal inferences about brain function. The distinct clinical profiles of semantic dementia, stroke aphasia, and ADHD provide compelling, real-world evidence for the models of semantic cognition and attentional control discussed previously, illustrating the profound consequences of disrupting the delicate interplay between representation and control.

Semantic Dementia vs. Stroke Aphasia: A Dissociation of Representation and Control

The comparison between semantic dementia (SD) and certain forms of stroke aphasia provides a classic double dissociation that serves as a cornerstone for the modern understanding of semantic cognition, particularly the Controlled Semantic Cognition (CSC) framework.⁵⁰ Although patients from both groups can present with poor comprehension and obtain broadly similar scores on some semantic tests, the underlying nature of their deficit is qualitatively different, reflecting damage to distinct components of the semantic system.

Semantic Dementia (SD): A Deficit of Representation. SD is caused by progressive, focal neurodegeneration of the anterior temporal lobes (ATLs), the putative hub of the semantic representation system.⁶ Consequently, the core deficit in SD is a "storage deficit"—the amodal conceptual knowledge itself degrades and is ultimately lost.⁵⁰ This leads to a highly characteristic pattern of impairment. The deficit is multimodal, affecting comprehension and production across words, pictures, sounds, and objects.⁵⁰ Performance is remarkably consistent; an item that is unknown on one occasion will be unknown on subsequent occasions, because the underlying representation is gone.⁵⁰ The breakdown of knowledge is highly sensitive to concept familiarity and frequency, with common, robustly represented concepts being preserved longer than rare ones.⁵⁰ Naming errors are typically coordinate (e.g., calling a "sparrow" a "bird") or superordinate (e.g., calling a "sparrow" an "animal"), reflecting the loss of fine-grained featural knowledge while broader categorical information is retained.⁵⁰ Crucially, because the knowledge is lost, patients derive little to no benefit from external cues (e.g., a phonemic cue for a word they cannot name).⁵¹

Semantic Aphasia (Post-Stroke): A Deficit of Control. In contrast, the syndrome of semantic aphasia is typically caused by stroke-induced damage to left-hemisphere

frontoparietal regions, such as the inferior frontal gyrus (IFG) or posterior temporoparietal cortex. These areas are core nodes of the semantic *control* network (SCN), while the ATL hub often remains structurally intact.⁵⁰ The resulting deficit is therefore an "access deficit." The conceptual knowledge is largely preserved but cannot be reliably and flexibly retrieved or applied to the task at hand.⁵⁴ This leads to a very different behavioral profile. Performance is highly inconsistent and exquisitely sensitive to task demands and context, but not to concept familiarity.⁵⁰ For example, patients may succeed at matching highly associated items (which require little control) but fail with weakly associated items (which require high control).⁵⁶ Unlike SD patients, they can benefit substantially from cues, which help to constrain the retrieval process and guide them to the correct, intact representation.⁵¹ A hallmark of this control deficit is a "refractory" pattern of performance, where accuracy declines across repeated presentations of related items, reflecting an inability to manage the buildup of semantic competition and interference from previously relevant concepts.⁵⁶

This stark dissociation, summarized in the table below, provides powerful causal evidence for the dual architecture of semantic cognition proposed by the CSC model. It demonstrates that the ability to use knowledge depends on at least two separable neural systems: one for storing the representations (ATL) and another for controlling their access (frontoparietal SCN).

Feature	Semantic Dementia (SD)	Semantic Aphasia (Post-Stroke)
Primary Lesion Site	Bilateral Anterior Temporal Lobes (ATL)	Left Inferior Frontal / Temporoparietal Cortex
Core Deficit Type	Representation / Storage (Degraded Knowledge)	Control / Access (Impaired Retrieval)
Performance Consistency	Highly consistent across time and tasks	Inconsistent; sensitive to task demands
Effect of Familiarity	Strong effect (frequent items preserved longer)	Minimal/no effect
Effect of Cues	Minimal benefit (knowledge is gone)	Substantial benefit (helps access intact knowledge)
Typical Naming Errors	Coordinate (e.g., "cat" for	Associative (e.g., "leash" for

	"dog"), Superordinate ("animal")	"dog"), No response
Repetition Effect	N/A (storage deficit)	Refractory (performance declines with repetition)

The broader context of aphasia also highlights the importance of attentional control. Attentional deficits are common following stroke and are not monolithic; they vary widely across patients and tasks, with performance on complex auditory attention tasks often being particularly impaired.⁵⁸ The executive control impairments seen in semantic aphasia are, in essence, a domain-specific form of attentional dysfunction. This underscores that a modern, neurocognitive approach to aphasia must move beyond simple descriptive labels (like "Wernicke's aphasia") and instead characterize deficits at the level of the underlying functional systems—representation, control, and basic attention—to achieve accurate diagnosis and develop targeted rehabilitation strategies.⁵⁰

Attention-Deficit/Hyperactivity Disorder (ADHD): Executive Dysfunction Meets Semantic Conflict

Attention-Deficit/Hyperactivity Disorder (ADHD) is a neurodevelopmental disorder defined by pervasive and impairing symptoms of inattention, hyperactivity, and impulsivity.⁶¹ At its core, ADHD is understood as a disorder of executive functions, which encompass the suite of top-down mental processes responsible for self-regulation, including attentional control, inhibition, and working memory.⁶² These fundamental deficits in control have direct and measurable consequences for how individuals with ADHD engage with their semantic knowledge.

Studies examining semantic processing in this population have found that adults with ADHD are significantly less efficient than neurotypical controls at resolving semantic conflict. For instance, when presented with tasks that pit the literal meaning of a sentence against a conflicting metaphorical meaning, individuals with ADHD struggle to manage the ambiguity and select the context-appropriate interpretation.⁶⁷ This difficulty is a direct manifestation of their underlying executive dysfunction; the impaired attentional control and inhibitory mechanisms make it harder to suppress the

irrelevant meaning and focus on the relevant one.

Intriguingly, the neurocognitive strategy used to resolve this conflict appears to be qualitatively different in individuals with ADHD. Whereas neurotypical individuals rely heavily on their executive attention capacity to manage semantic conflict, the performance of individuals with ADHD is more strongly correlated with their semantic fluency—that is, the ease and speed with which they can retrieve words from their semantic store.⁶⁷ This suggests a compensatory mechanism: faced with less reliable top-down control, they may rely more on the bottom-up strength and accessibility of representations within their semantic network.

This observation points to a deeper, developmental relationship between attention and semantics. The chronic, lifelong deficits in attentional control that characterize ADHD may not only affect how semantic memory is *used* in the moment, but also how it is *built* over a lifetime. There is evidence suggesting that the semantic networks of children with ADHD may be structured differently, showing weaker priming effects for abstract, low-association word pairs.⁶⁸ Forming such links requires more cognitive effort and sustained, controlled attention to integrate disparate pieces of information. A lifetime of difficulty in sustaining this kind of controlled cognitive engagement could lead to a semantic network that is more sparsely connected in its more abstract domains. This developmental cascade—where an early deficit in a foundational process like attention shapes the subsequent development of another cognitive system like semantic memory—argues against a simple view of attention as a concurrent support process. Instead, it suggests that attentional control is a fundamental mechanism that actively sculpts the structure of our knowledge of the world over our entire lifespan.

Part V: Synthesis, Emerging Theories, and Future Directions

The journey through the neural architectures of semantic cognition and attentional control, their dynamic interplay, and their dissolution in clinical syndromes culminates in a unified, network-based understanding of human thought. This final section synthesizes the core findings of this report, highlights the major unresolved debates that animate the field, and looks toward the methodological and theoretical frontiers that will shape the future of cognitive neuroscience.

A Unified View: The Codependence of Semantic and Attentional Networks

The central thesis emerging from the wealth of evidence reviewed is that semantic cognition and attention are not independent faculties that simply interact. They are fundamentally codependent, emergent properties of a dynamic, recursive relationship between large-scale brain networks. Flexible, goal-directed, and meaningful behavior arises from the brain's remarkable ability to continuously couple and decouple its core functional systems. These systems include: a stable yet plastic **semantic representation system** (the hub-and-spoke network) that houses our world knowledge; a specialized **semantic control network (SCN)** that flexibly guides retrieval from this knowledge store; and **domain-general attentional networks** (the DAN and VAN) that manage the interface between internal goals and external stimuli. All of these operations unfold within the high-level associative workspace provided by the brain's **Default Mode Network (DMN)**. Intelligence, in this view, is not located in any single node or network, but in the brain's capacity for context-sensitive network reconfiguration—the ability to form the right functional coalitions at the right time to meet the demands of the moment.

Current Debates and Unanswered Questions

Despite significant progress, several fundamental questions remain at the forefront of research, representing key challenges and opportunities for the field.

First, **the enigma of the angular gyrus (AG)** persists. The AG is a major convergence zone in the brain, and it is consistently activated in semantic paradigms and implicated as a core node of the DMN. Yet, its precise contribution to semantic cognition remains a subject of intense debate. Some models propose it is a high-level semantic hub, akin to the ATL. However, its functional profile challenges this view. The AG often shows deactivation relative to rest, its activity scales with general task difficulty rather than specific semantic content, and it is often anti-correlated with the dorsal attention network.³ These characteristics have led to alternative theories proposing a more domain-general function, such as an online buffer for incoming multimodal information, a key component of the episodic memory retrieval system, or a central node for the DMN's role in internal mentation. Resolving the true

computational function of the AG is a critical step toward a complete model of the semantic-attention interface.

Second, the question of **the functional coherence of the DMN** is central. The evidence presented strongly suggests that the DMN is not a unified system with a single purpose. Instead, it is better conceptualized as a "dynamic patchwork" of functionally distinct but interacting subsystems.⁴⁵ Different components of the DMN show unique response profiles to semantic content, input modality, and task demands. Future research must therefore move beyond treating the DMN as a monolithic entity. The key will be to investigate the specific computations performed by its constituent parts (e.g., the core, medial temporal, and frontotemporal subsystems) and to model how their interactions give rise to the diverse cognitive states—from semantic processing to autobiographical recall—with which the network is associated.

Third, **the nature of cognitive control** requires further clarification. The discovery of a semantic-specific control network (SCN) that is at least partially dissociable from the domain-general Multiple-Demand Network (MDN) raises fundamental questions about how the brain implements control.⁴² How do these two control systems divide labor? Do they operate in parallel or in a hierarchical fashion? Is the SCN simply a specialized instantiation of the MDN, or does it rely on fundamentally different computational principles? Understanding how the brain achieves both general-purpose and domain-specific cognitive control is a crucial frontier in neuroscience.

Methodological and Theoretical Frontiers

The path forward in addressing these complex questions will be driven by advances in both methodology and theory. The field is in the midst of a paradigm shift, moving away from a simple localizationist approach toward a dynamic, network-based view of cognition.

Multimodal Neuroimaging: No single imaging technique can capture the full spatiotemporal complexity of brain function. The future lies in the sophisticated integration of multiple methods. By combining the high spatial resolution of **fMRI**, the exceptional temporal resolution of **EEG and MEG**, and the detailed structural connectivity information from **DTI**, researchers can build a much more complete

picture of neurocognitive events.³⁷ This multimodal approach is essential for linking function to anatomy and for dissociating the rapid, automatic stages of processing from the slower, more controlled and attention-dependent stages. Novel techniques like functional near-infrared spectroscopy (fNIRS) and ultra-high field fMRI will continue to push these boundaries.⁶⁹

Computational and AI Models: To move from describing brain activity to explaining the mechanisms that produce it, formal computational modeling is indispensable. Frameworks like the Controlled Semantic Cognition model provide concrete, testable hypotheses about the interactions between system components.¹⁰ A particularly powerful driver of future progress is the growing synergy between cognitive neuroscience and artificial intelligence. The development of "attention mechanisms" in AI, initially inspired by neuroscience, has revolutionized machine learning.¹⁶ Now, more sophisticated "semantic attention" mechanisms are being developed, which provide novel, implemented theories about how a system can achieve context-aware processing.⁷³ This creates a virtuous cycle: AI provides formal, testable models of cognition, while neuroscience provides the biological ground truth and architectural inspiration for building more robust and human-like artificial intelligence.

Cognitive Network Neuroscience: Ultimately, the future of the field lies in the full embrace of a network science perspective. This approach, often termed "human connectomics," treats the brain not as a collection of independent regions, but as a single, integrated, and complex system.⁶ The key questions are no longer just "where" a function happens, but "how" it emerges from the topological properties of the network and its dynamic reconfigurations across different cognitive states. Resolving the debates surrounding hubs like the AG and networks like the DMN will require precisely these kinds of approaches—methods that can characterize how a region's pattern of functional connectivity and its position within the global network architecture change as a function of the cognitive context. The function of a hub may not be a static property, but rather the capacity to facilitate these dynamic reconfigurations. By modeling the principles of network interaction, dynamic coupling, and state transitions, we can hope to finally crack the code of the complex, emergent phenomena that we call attention and semantics.

Works cited

1. The semantic anatomical network: Evidence from healthy and brain ..., accessed August 11, 2025, <https://pmc.ncbi.nlm.nih.gov/articles/PMC6869673/>
2. Elucidating a Core Semantic Network in the Brain—Implications for ..., accessed August 11, 2025, <https://pmc.ncbi.nlm.nih.gov/articles/PMC6604887/>
3. Functional connectivity of brain networks during semantic processing in older

adults, accessed August 11, 2025,

<https://www.frontiersin.org/journals/aging-neuroscience/articles/10.3389/fnagi.2022.814882/full>

4. The Semantic Network at Work and Rest: Differential Connectivity of Anterior Temporal Lobe Subregions - PubMed Central, accessed August 11, 2025, <https://pmc.ncbi.nlm.nih.gov/articles/PMC4737765/>
5. The Brain's Semantic Network - Number Analytics, accessed August 11, 2025, <https://www.numberanalytics.com/blog/brain-semantic-network-dynamic-semantics-language>
6. The Brain Connectivity Basis of Semantic Dementia: A Selective ..., accessed August 11, 2025, <https://pmc.ncbi.nlm.nih.gov/articles/PMC6493106/>
7. Brain Network for the Core Deficits of Semantic Dementia: A Neural Network Connectivity-Behavior Mapping Study - Frontiers, accessed August 11, 2025, <https://www.frontiersin.org/journals/human-neuroscience/articles/10.3389/fnhum.2017.00267/full>
8. The brain network in support of social semantic accumulation - PMC, accessed August 11, 2025, <https://pmc.ncbi.nlm.nih.gov/articles/PMC7990071/>
9. Language networks in semantic dementia | Brain - Oxford Academic, accessed August 11, 2025, <https://academic.oup.com/brain/article/133/1/286/308115>
10. Controlled semantic cognition relies upon dynamic and flexible interactions between the executive 'semantic control' and hub-and-spoke 'semantic representation' systems - ResearchGate, accessed August 11, 2025, https://www.researchgate.net/publication/323673501_Controlled_semantic_cognition_relies_upon_dynamic_and_flexible_interactions_between_the_executive_'semantic_control'_and_hub-and-spoke_'semantic_representation'_systems
11. Controlled semantic cognition relies upon dynamic and flexible ..., accessed August 11, 2025, <https://pmc.ncbi.nlm.nih.gov/articles/PMC6006425/>
12. The Brain Connectivity Basis of Semantic Dementia: A Selective Review - PubMed, accessed August 11, 2025, <https://pubmed.ncbi.nlm.nih.gov/26336932/>
13. The organization of the semantic network as reflected by the neural correlates of six semantic dimensions - PubMed, accessed August 11, 2025, <https://pubmed.ncbi.nlm.nih.gov/38295716/>
14. The brain network in support of social semantic accumulation - PubMed, accessed August 11, 2025, <https://pubmed.ncbi.nlm.nih.gov/33433627/>
15. The Neurobiology of Attention - Number Analytics, accessed August 11, 2025, <https://www.numberanalytics.com/blog/neurobiology-attention-developmental-processes-mechanisms>
16. Attention in Psychology, Neuroscience, and Machine ... - Frontiers, accessed August 11, 2025, <https://www.frontiersin.org/journals/computational-neuroscience/articles/10.3389/fncom.2020.00029/full>
17. From behavior to neural dynamics: An integrated theory of attention - PMC - PubMed Central, accessed August 11, 2025, <https://pmc.ncbi.nlm.nih.gov/articles/PMC4604109/>
18. (PDF) Neurobiology of Attention Deficit Hyperactivity Disorder (ADHD): A brief

- review, accessed August 11, 2025,
https://www.researchgate.net/publication/378857889_Neurobiology_of_Attention_Deficit_Hyperactivity_Disorder_ADHD_A_brief_review
19. Attention can be subdivided into neurobiological components ..., accessed August 11, 2025, <https://www.pnas.org/doi/10.1073/pnas.1902286116>
 20. Asymmetric Development of Dorsal and Ventral Attention Networks in the Human Brain, accessed August 11, 2025,
https://www.researchgate.net/publication/272200605_Asymmetric_Development_of_Dorsal_and_Ventral_Attention_Networks_in_the_Human_Brain
 21. Involvement of the dorsal and ventral attention networks in visual attention span - PMC, accessed August 11, 2025,
<https://pmc.ncbi.nlm.nih.gov/articles/PMC8933248/>
 22. Top-down modulation of attention by emotion - Frontiers, accessed August 11, 2025,
<https://www.frontiersin.org/journals/human-neuroscience/articles/10.3389/fnhum.2013.00102/full>
 23. (PDF) Dorsal and Ventral Attention Systems: Distinct Neural Circuits but Collaborative Roles, accessed August 11, 2025,
https://www.researchgate.net/publication/247771573_Dorsal_and_Ventral_Attention_Systems_Distinct_Neural_Circuits_but_Collaborative_Roles
 24. Dorsal and Ventral Attention Systems: Distinct Neural Circuits but ..., accessed August 11, 2025, <https://pmc.ncbi.nlm.nih.gov/articles/PMC4107817/>
 25. Spontaneous neuronal activity distinguishes human dorsal and ventral attention systems, accessed August 11, 2025,
<https://www.pnas.org/doi/10.1073/pnas.0604187103>
 26. Neurobiology of attention-deficit hyperactivity disorder: historical challenges and emerging frontiers - Oregon Health & Science University, accessed August 11, 2025,
<https://ohsu.elsevierpure.com/en/publications/neurobiology-of-attention-deficit-hyperactivity-disorder-historic>
 27. Neurobiology of attention-deficit hyperactivity disorder: historical challenges and emerging frontiers - Experts@Minnesota, accessed August 11, 2025,
<https://experts.umn.edu/en/publications/neurobiology-of-attention-deficit-hyperactivity-disorder-historic>
 28. Neuroimaging Studies of Attention: From Modulation of Sensory Processing to Top-Down Control - PMC - PubMed Central, accessed August 11, 2025,
<https://pmc.ncbi.nlm.nih.gov/articles/PMC6741071/>
 29. Top-down modulation: Bridging selective attention and working memory - PubMed Central, accessed August 11, 2025,
<https://pmc.ncbi.nlm.nih.gov/articles/PMC3510782/>
 30. CHAPTER TWO - Attentional control of lexical processing pathways during word recognition and reading, accessed August 11, 2025,
<http://psychnet.wustl.edu/coglab/wp-content/uploads/2015/01/Attentional-control-of-lex-process-1999.pdf>
 31. Interactions of Top-Down and Bottom-Up Mechanisms in Human Visual Cortex,

- accessed August 11, 2025, <https://www.jneurosci.org/content/31/2/587>
32. An fMRI Study of Response and Semantic Conflict in the ... - Frontiers, accessed August 11, 2025, <https://www.frontiersin.org/journals/psychology/articles/10.3389/fpsyg.2019.02426/full>
 33. Attentional constraints on semantic activation: Evidence from Stroop's paradigm | Request PDF - ResearchGate, accessed August 11, 2025, https://www.researchgate.net/publication/307884055_Attentional_constraints_on_semantic_activation_Evidence_from_Stroop's_paradigm
 34. The semantic Stroop effect is controlled by endogenous attention - Apollo, accessed August 11, 2025, <https://www.repository.cam.ac.uk/items/09a218e9-bb7d-40fe-ad61-62ce490230d6>
 35. Semantic and attentional networks in bilingual processing: fMRI connectivity signatures of translation directionality - PMC, accessed August 11, 2025, <https://pmc.ncbi.nlm.nih.gov/articles/PMC7933822/>
 36. Attentional Control and the Relatedness Proportion Effect in Semantic Priming, accessed August 11, 2025, https://www.researchgate.net/publication/6261281_Attentional_Control_and_the_Relatedness_Proportion_Effect_in_Semantic_Priming
 37. Automatic Semantic Facilitation in Anterior Temporal Cortex ..., accessed August 11, 2025, <https://www.jneurosci.org/content/33/43/17174>
 38. Guidance of visual attention by semantic information in real-world ..., accessed August 11, 2025, <https://pmc.ncbi.nlm.nih.gov/articles/PMC3915098/>
 39. Semantic guidance of attention within natural scenes - University of Oregon, accessed August 11, 2025, https://pages.uoregon.edu/prd/pdfs/Walter_Dassonville_VisCog05.pdf
 40. Meaning guides attention during scene viewing, even when it is irrelevant - eScholarship, accessed August 11, 2025, <https://escholarship.org/uc/item/264393sh>
 41. Full article: The structure of semantic representation shapes controlled semantic retrieval - Taylor & Francis Online, accessed August 11, 2025, <https://www.tandfonline.com/doi/full/10.1080/09658211.2021.1906905>
 42. Flexible semantic cognition - University of York, accessed August 11, 2025, <https://www.york.ac.uk/biomedical-research-institute/neuroscience/flexible-semantic-cognition/>
 43. Macroscale brain states support the control of semantic cognition - PMC - PubMed Central, accessed August 11, 2025, <https://pmc.ncbi.nlm.nih.gov/articles/PMC11294576/>
 44. Role of the Default Mode Network in Cognitive Transitions | Cerebral Cortex | Oxford Academic, accessed August 11, 2025, <https://academic.oup.com/cercor/article/28/10/3685/5060278>
 45. Distinctive and Complementary Roles of Default Mode Network Subsystems in Semantic Cognition | Journal of Neuroscience, accessed August 11, 2025, <https://www.jneurosci.org/content/44/20/e1907232024>

46. Self-Processing and the Default Mode Network: Interactions with the Mirror Neuron System - Frontiers, accessed August 11, 2025, <https://www.frontiersin.org/journals/human-neuroscience/articles/10.3389/fnhum.2013.00571/full>
47. Portraying the unique contribution of the default mode network to internally driven mnemonic processes | PNAS, accessed August 11, 2025, <https://www.pnas.org/doi/10.1073/pnas.1209888110>
48. Establishing task- and modality-dependent dissociations between ..., accessed August 11, 2025, <https://www.pnas.org/doi/10.1073/pnas.1422760112>
49. A default mode network subsystem supports both item and associative word encoding: Insights from a meta-analysis | Imaging Neuroscience - MIT Press Direct, accessed August 11, 2025, https://direct.mit.edu/imag/article/doi/10.1162/imag_a_00321/124780/A-default-m-ode-network-subsystem-supports-both
50. Semantic impairment in stroke aphasia versus semantic dementia: a case-series comparison | Brain | Oxford Academic, accessed August 11, 2025, <https://academic.oup.com/brain/article/129/8/2132/333526>
51. Semantic impairment in stroke aphasia versus semantic dementia: a case-series comparison - Research Explorer, accessed August 11, 2025, https://research.manchester.ac.uk/files/21742146/POST-PEER-REVIEW-PUBLISHER_S.PDF
52. Semantic impairment in aphasia: a problem of control? - Macquarie University, accessed August 11, 2025, <https://researchers.mq.edu.au/en/publications/semantic-impairment-in-aphasia-a-problem-of-control>
53. Episodic memory in frontotemporal dementia: a critical review | Brain - Oxford Academic, accessed August 11, 2025, <https://academic.oup.com/brain/article/135/3/678/1746798>
54. Semantic Processing in Bilingual Aphasia: Evidence of ... - Frontiers, accessed August 11, 2025, <https://www.frontiersin.org/journals/human-neuroscience/articles/10.3389/fnhum.2019.00205/full>
55. Semantic dementia: a complex and culturally influenced presentation - PMC, accessed August 11, 2025, <https://pmc.ncbi.nlm.nih.gov/articles/PMC10801360/>
56. Varieties of semantic 'access' deficit in Wernicke's aphasia and ..., accessed August 11, 2025, <https://academic.oup.com/brain/article/138/12/3776/412124>
57. Semantic impairment in aphasia: A problem of control? | Request PDF - ResearchGate, accessed August 11, 2025, https://www.researchgate.net/publication/364374580_Semantic_impairment_in_a_phasis_A_problem_of_control
58. Attention to attention in aphasia – elucidating impairment patterns, modality differences and neural correlates - PMC, accessed August 11, 2025, <https://pmc.ncbi.nlm.nih.gov/articles/PMC7614452/>
59. Is there an impairment of language-specific attention processing in aphasia? | Request PDF, accessed August 11, 2025,

- https://www.researchgate.net/publication/247303382_Is_there_an_impairment_of_language-specific_attention_processing_in_aphasia
60. Attention and Other Cognitive Deficits in Aphasia: Presence and Relation to Language and Communication Measures - ASHA Journals, accessed August 11, 2025, <https://pubs.asha.org/doi/10.1044/1058-0360%282012/11-0067%29>
 61. Attention deficit hyperactivity disorder - Wikipedia, accessed August 11, 2025, https://en.wikipedia.org/wiki/Attention_deficit_hyperactivity_disorder
 62. Attentional Control Activation Relates to Working Memory in Attention-Deficit/Hyperactivity Disorder - PMC - PubMed Central, accessed August 11, 2025, <https://pmc.ncbi.nlm.nih.gov/articles/PMC2953472/>
 63. The study of automatic and controlled processes in ADHD: A reread and a new proposal, accessed August 11, 2025, https://www.researchgate.net/publication/316786239_The_study_of_automatic_and_controlled_processes_in_ADHD_A_reread_and_a_new_proposal
 64. What Is Executive Function? 7 Deficits Tied to ADHD - ADDitude, accessed August 11, 2025, <https://www.additudemag.com/7-executive-function-deficits-linked-to-adhd/>
 65. Executive Function in Children with Attention Deficit/Hyperactivity ..., accessed August 11, 2025, <https://pmc.ncbi.nlm.nih.gov/articles/PMC4425416/>
 66. Executive Function and Attention Performance in Children with ADHD: Effects of Medication and Comparison with Typically Developing Children - MDPI, accessed August 11, 2025, <https://www.mdpi.com/1660-4601/16/20/3822>
 67. Semantic conflicts are resolved differently by adults with and without ..., accessed August 11, 2025, <https://pubmed.ncbi.nlm.nih.gov/26513742/>
 68. The structure and function of semantic memory in children with attention-deficit/hyperactivity disorder - University of Texas at Austin, accessed August 11, 2025, <https://repositories.lib.utexas.edu/items/2490f843-04e1-4a9c-9df5-89148a893fc0>
 69. Exploring the Frontiers of Neuroimaging: A Review of Recent Advances in Understanding Brain Functioning and Disorders - MDPI, accessed August 11, 2025, <https://www.mdpi.com/2075-1729/13/7/1472>
 70. pmc.ncbi.nlm.nih.gov, accessed August 11, 2025, <https://pmc.ncbi.nlm.nih.gov/articles/PMC11336413/#:~:text=Neuroimaging%20techniques%2C%20such%20as%20electroencephalography,the%20human%20brain%20network%20system.>
 71. Late maturation of semantic control promotes conceptual development - eLife, accessed August 11, 2025, <https://elifesciences.org/reviewed-preprints/105310>
 72. Top-Down Visual Attention From Analysis by Synthesis - CVF Open Access, accessed August 11, 2025, https://openaccess.thecvf.com/content/CVPR2023/papers/Shi_Top-Down_Visual_Attention_From_Analysis_by_Synthesis_CVPR_2023_paper.pdf
 73. Semantic Attention Guided Hierarchical Decision Network for Age Estimation - MDPI, accessed August 11, 2025, <https://www.mdpi.com/2076-3417/15/14/7707>
 74. The cognitive impacts of large language model interactions on problem solving

and decision making using EEG analysis - Frontiers, accessed August 11, 2025,
<https://www.frontiersin.org/journals/computational-neuroscience/articles/10.3389/fncom.2025.1556483/full>

75. Cognitive Network Neuroscience - PMC - PubMed Central, accessed August 11, 2025, <https://pmc.ncbi.nlm.nih.gov/articles/PMC4854276/>